

# Complex Analysis – 7

## Конформные отображения и дробно-линейные преобразования

1 сентября 2026 года

### Теоретический минимум.

**Опр.** A Möbius transformation is  $T(z) = \frac{az+b}{cz+d}$  with  $ad-bc \neq 0$ ; these are exactly the automorphisms of the Riemann sphere. The *cross-ratio*  $(z_1, z_2; z_3, z_4)$  is preserved by every Möbius map, and Möbius maps take circles and lines to circles and lines.

**Теорема 1 (теорема Римана об отображении)** *Every simply connected domain  $\Omega \subsetneq \mathbb{C}$  admits a biholomorphic map onto the unit disc; it becomes unique once we prescribe  $f(z_0) = 0$  and  $f'(z_0) > 0$ .*

**Теорема 2 (принцип симметрии Шварца)** *If  $f$  is holomorphic on  $\Omega^+$  (a domain in the upper half-plane) and extends continuously to a segment  $I$  of the real axis with real values there, then  $f(\bar{z}) = \overline{f(z)}$  defines a holomorphic extension of  $f$  across  $I$ .*

**Факт 1** Стандартный набор:  $z \mapsto \frac{z-i}{z+i}$  переводит верхнюю полуплоскость в круг,  $z \mapsto z^\alpha$  раскрывает угол,  $z \mapsto e^z$  переводит полосу в полуплоскость,  $z \mapsto \frac{1}{2}(z + \frac{1}{z})$  (Жуковский) переводит внешность круга в дополнение к отрезку.

---

1. Prove that a Möbius transformation preserves the cross-ratio and maps circles-and-lines to circles-and-lines; then find the map of the upper half-plane onto  $\mathbb{D}$  sending  $i$  to 0, and describe all automorphisms of the upper half-plane.
2. (S.T. Yau CSMC 2012) Construct a one-to-one conformal mapping from the lune  $U = \{z : |z - \frac{i}{2}| < \frac{1}{2}\} \setminus \{z : |z - \frac{i}{4}| \leq \frac{1}{4}\}$  onto the unit disc.

3. (S.T. Yau CSMC 2017) Find a conformal mapping of the lens-shaped region  $\{z : |z - i| < \sqrt{2}\} \cap \{z : |z + i| < \sqrt{2}\}$  onto the unit disc.
4. (S.T. Yau CSMC 2014) Prove that if there is a conformal mapping between the annuli  $\{r_1 < |z| < r_2\}$  and  $\{\rho_1 < |z| < \rho_2\}$ , then  $\frac{r_2}{r_1} = \frac{\rho_2}{\rho_1}$ .
5. (S.T. Yau CSMC 2017) Prove that  $w \mapsto z = \int_0^w (1 - x^n)^{-2/n} dx$  is a conformal mapping of the unit disc onto the interior of a regular  $n$ -gon  $\Gamma$ , and write down the solution of the Dirichlet problem  $\Delta u = 0$ ,  $u|_{\partial\Gamma} = \varphi$  in that polygon.